

Midterm 1 Concept Inventory

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This document is subject to change till February 20, 11:59pm, after which only “Exclusions” will grow.

1 Overview

Midterm 1 covers sections I and II in Trefethen Bau and our study of the Discrete Fourier Transform and FFT algorithm. The midterm may involve computation, but only (near-) minimum cases to illustrate the mathematics, not giant computations better suited for computation. Generally, that means 2-by-2 or 3-by-3 matrices, but also large sparse matrices (mostly zeros). For the midterm, you’ll need to describe algorithms in English and pseudocode, but not do any coding that should be done with the feedback of a computer. Simple proofs are fair game. Geometric and symbol-pushing views and their connections. We’ve seen some examples involving stability, but we haven’t yet started studying it thoroughly.

2 Details

1. Matric-vector multiplication. In AB , each column of B picks out a linear combination of columns of A . That includes $B = v$, a vector. Each row of A picks out a linear combination of rows of B .
2. Vandermonde matrices: $V\vec{c} = \vec{y}$ says that a polynomial with coefficient sequence $\vec{c}$, when evaluated at points in column number 1 of V (starting with the all-1’s column as number zero), gives the values in $\vec{y}$. Fourier F and F^* are examples.
3. Basics: range, nullspace, determinant, inverse, rank, transpose, adjoint (conjugate-transpose), inner products, outer products, etc.
4. Basic vocabulary is fair game. (But the corner cases and fine points are excluded, particularly when authors differ on details. Such details will be provided.) SVD and QR are considered fundamental vocabulary. “Full” versus “Reduced” warrants a reminder.
5. Orthogonal vectors, resolution of a vector into orthogonal components, unitary matrices, Multiplying a unitary matrix by a vector as change of bases, dot products, projection: for vector v and unit vector u , the expressions v onto scalar u^*v or vector $u(u^*v)$, which scales u by u^*v , and $(uu^*)v = u(u^*v)$, in which the outer product uu^* projects v orthogonally onto the span of u .
6. Kroneker product and Hadamard
7. Preservation and transformation of properties: $(AB)^{-1} = B^{-1}A^{-1}$, Kronecker product of unitary is unitary, etc.
8. p -norms of vectors, with intuition for $p = 1, 2, \infty$. Unit balls under these norms. Induced matrix norms. Computing 1- and ∞ - norms of matrices. Working with 2-norm of matrices (e.g., computing from singular value decomposition (which we don’t yet know how to compute) as the largest singular value. Simple bounds on norms, e.g., $\max(M) \leq \|M\|_1, \|M\|_\infty \leq \max(m, n) \max(M), \|AB\| \leq \|A\| \cdot \|B\|$,

norms of structured matrices (diagonal, rank-1), Frobenius norm. Unitary transformations preserve 2-norm and Frobenius, Cauchy-Schwarz

9. SVD: computing for 2-by-2 matrices (but not bigger, unless the matrix is structured). Geometric interpretations (semi-major and semi-minor axes of ellipses; rotate-scale along axes-rotate), full and reduced, orthogonal to orthogonal bases, low-rank approximations, reading determinant, etc., from the SVD. Solving systems and least squares
10. orthogonal projections: changing some 1's to 0 in the identity matrix in a natural basis, QQ^* for orthogonal Q , complementary projections
11. QR factorizations: from classic Gram-Schmidt, modified Gram-Schmidt, and Householder, orthogonal polynomials, solving $Ax = b$ and solving least squares from the form $QRx = b$
12. Projections and Reflections, and in their natural bases
13. Least squares: Minimize $\|Ax - b\|_2$ when b is not in the range of A . Geometric view, with right angle at closest point Ax_* : the vector $b - Ax_*$ is normal to the hyperplane $\text{range}(A)$. By normal equations, QR factorization, and SVD factorization. Normal equations $A^*(Ax - b) = 0$ tests $Ax - b$ for zero by the columns of A (rows of A^*), and a non-zero result gives a way to change x to improve Ax . Equivalent form $x = (A^*A)^{-1}A^*b$, whence $Ax = A(A^*A)^{-1}A^*b$ is the closest point to b in the range of A
14. Computational cost in $O()$ form (not caring about constant factor) for algorithms we've seen (including FFT). For using QR or SVD, we are interested not only in the cost to do the factorization, but also **after having** the factorization. For example, if $A = QR$ and we know Q and R we can reuse the factorization to solve $Ax = b_1, Ax = b_2, \dots$, for the same A as b changes.
15. Complex numbers: $re^{i\theta} = r \cos(\theta) + ir \sin(\theta)$, roots of unity, complex conjugates $\overline{re^{i\theta}} = re^{-i\theta}$ and $\overline{a + bi} = a - bi$
16. FFT: Algorithm for $F_6 = F_2 \otimes F_3$, etc., using Kronecker product of Fourier of relatively-prime order (but not analysis of correctness or runtime beyond $n \log^{O(1)}(n)$), FFT of length a power of 2, via polynomial interpolation at roots of unity: recurse on even and odd roots, twiddle the odds, combine with $\frac{1 \pm z^{n/2}}{2}$; F is Vandermonde, so F and $F^* = F^{-1}$ do evaluation and interpolation, F^* reverses most rows versus F , for convolution: to get $r(z) = p(z)q(z)$, evaluate $p(z)$ and $q(z)$ at sufficiently-many points, multiply the evaluations, then interpolate to get coefficients of $r(z)$, Fourier diagonalizes the cyclic shift operator S , and so polynomials in S , which covers Circulant matrices (and, after hacks, Toeplitz matrices), which include differential operators and convolutions. That is, columns of Fourier are eigenvectors of all Circulants, including $S^0 = I$, and the corresponding eigenvalues of S are the roots of unity. If λ is an eigenvalue of S , the $p(\lambda)$ is the corresponding eigenvalue of the Circulant $p(S)$.
17. Everything on or adjacent to the sample exam

3 Exclusions (from Midterm, but still good stuff!)

1. Calculations and computations beyond (near-) minimum to illustrate Math principles
2. Proof of correctness and efficiency of the FFT of the form $F_6 = F_2 \otimes F_3$ and the like—the Kronecker product of relatively prime orders.
3. Multi-dimensional FFT
4. Full analysis of stability
5. Advanced vocabulary is off (though basic terms like vector and matrix are fair game). Minus signs and factors of 2, etc., will be provided, especially where the literature varies on definitions.